

CONTACT

- thisisnauchno@gmail.com
- +7 921 945 67 03
- Saint Petersburg, Russia
- GitHub
- Portfolio website
- ORCID
- eLibrary profile

SKILLS

ML / DS
Python, PyTorch, TensorFlow, Ultralytics YOLO, FastAPI, Scikit-learn, Scikit-image, NumPy, SciPy, Pandas, Matplotlib, Seaborn, LangChain, OpenAI API, Hugging Face, MLflow, Apache Spark (basics), Computer Vision

DevOps / CI-CD
Docker, Kubernetes, Terraform, Ansible, Jenkins

Databases
PostgreSQL, MySQL, MongoDB, Redis

Brokers
RabbitMQ

Maths & Theory
Statistical ML, Probability Theory, Mathematical Statistics, Optimization Methods, Algorithms & Data Structures, Discrete Mathematics, Quantum Computing

Other Langs / Frameworks
Haskell, C++ (basics), Java (basics), Spring (basics)

LANGUAGES

Russian
Native

English
B2

Azerbaijani
Intermediate

INTERESTS

Computer Vision, Mathematical Modeling in AI, Travel, Football, Music

Ayzek Salimli

Data Scientist

Professional Summary

Data Scientist with a strong mathematical and statistical background. First-year Master's student in *Machine Learning and Artificial Intelligence* at SPbPU, holding a Bachelor's degree in Mathematics and Computer Science (AI Systems and Supercomputer Technologies). Practical experience developing computer-vision and deep-learning models (PyTorch, TensorFlow, Ultralytics YOLO), integrating LLM solutions (LangChain + OpenAI API), solving vehicle-routing problems (VRP/DVRP) with optimization methods, and building production systems on FastAPI. Proficient with the DevOps stack (Docker, Kubernetes, Terraform, Ansible, Jenkins). Projects and publications in detail: portfolio website.

Work Experience

ML Engineer 08/2024 - 02/2026
TheBloomsBridge

- Integration of AI systems, including computer vision and agentic workflows, into the project using RAG, YOLO, YOLO-seg, SegFormer, OpenCV, LangChain, and LangGraph.
- Development of a project dashboard and algorithms for dynamic vehicle routing problem (dVRP) tasks.
- Research and training of semantic and instance segmentation models for real-time segmentation tasks.
- Development of an automated system for reconciling customer orders with the items collected by couriers.
- Development of a RAG system that provides recommendations based on reports.
- Clustering of users by their orders and geographic areas.
- Development of a zero-shot recommendation system for finding similar products.
- Forecasting the number of products required for procurement.
- Statistical and time-series analysis using ARIMA, SARIMA, and SARIMAX.
- Development of a dynamic route optimization algorithm accounting for the courier's mode of transport, road events, product characteristics, and weather conditions.
- Research into replacing the ESA attention mechanism with state-space models (SSMs) in ViT-style segmenters.

Mathematics Tutor 04/2023 - 05/2024
Private tutoring

- Preparation for state exams in mathematics.
- Linear algebra for applicants to technical degree programs.

Education

Master's Degree - Mathematics and Computer Science 07/2026 - 07/2028
SPbPU, ICCS, Graduate School of AI Technologies

- Program: Machine Learning and Artificial Intelligence.

Bachelor's Degree - Mathematics and Computer Science 09/2022 - 07/2026
SPbPU, ICCS, Graduate School of AI Technologies

- Program: Artificial Intelligence Systems and Supercomputer Technologies (AI and HPC).
- Thesis: algorithm for detecting foreign objects on tram tracks based on their segmentation. (Investigated and trained YOLO-seg v8/11/26 (n, s), SegFormer-B0 and DeepLabV3+ (ResNet-101) models; built a real-time pipeline with track segmentation, ego-track tracking, a dynamic RoI, YOLO detection and ByteTrack object tracking.)

Publications

- Application of Computer Vision Methods for Determining Offside Position in Football.
- Construction of a Region of Interest Based on Segmentation - Proceedings of the Science Week, p. 386 *(1st degree laureate)*.
- Improving the SegFormer Decoder for Semantic Segmentation by Adding a Depth-wise Convolution Block - Proceedings of the XXIII International Scientific and Practical Conference "Young Scientist", p. 52 *(1st place in the "Technical Sciences" section)*.